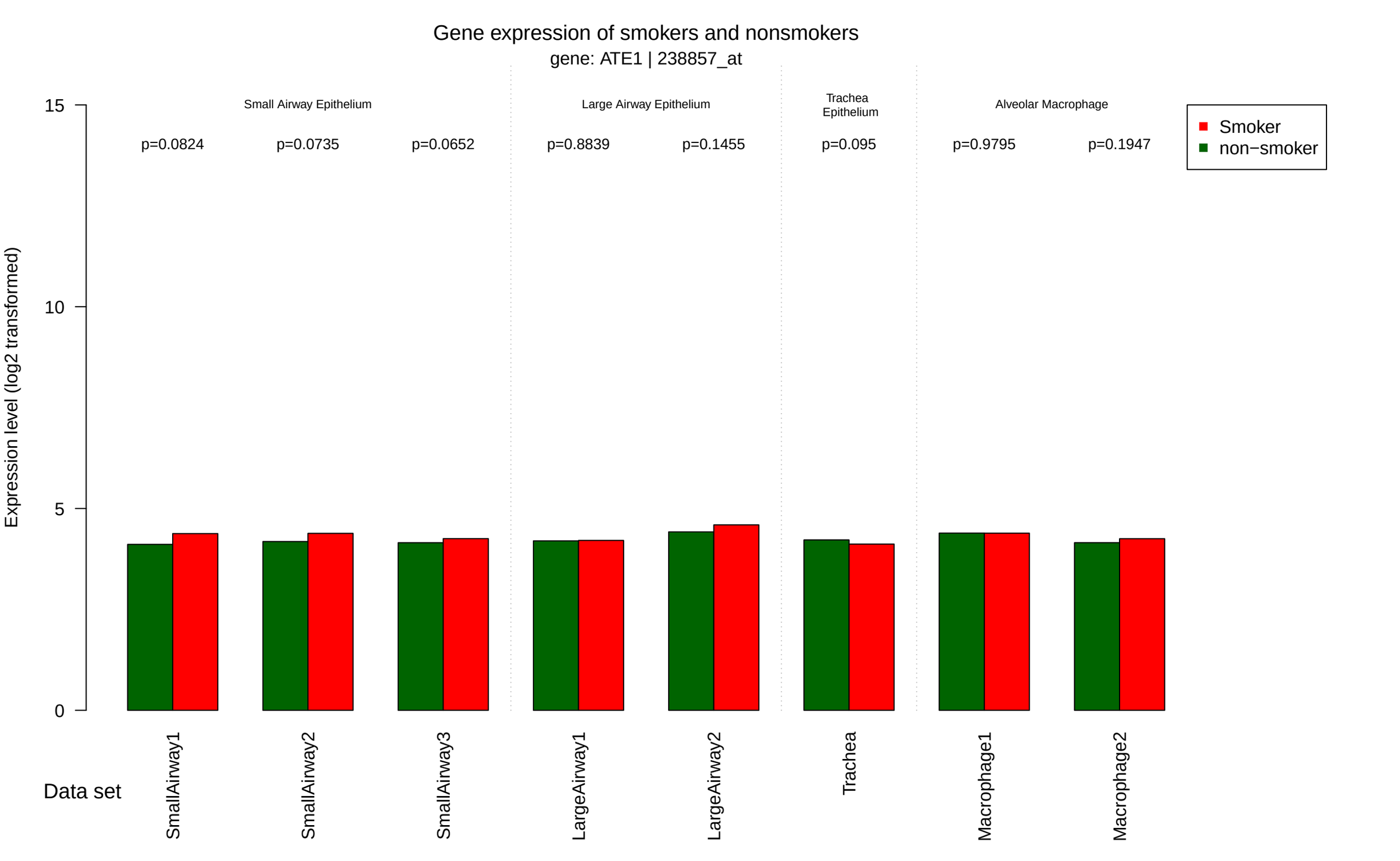
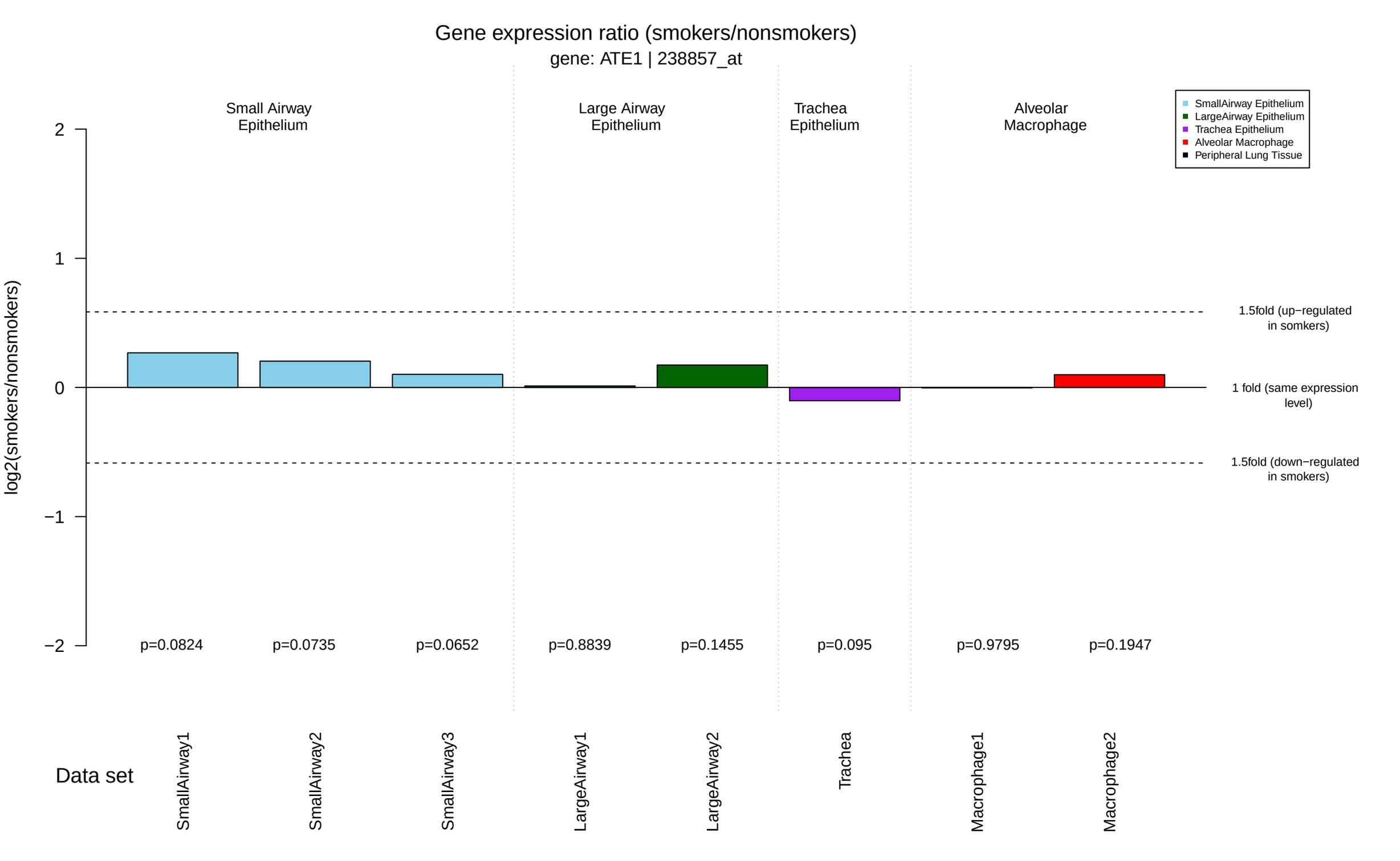
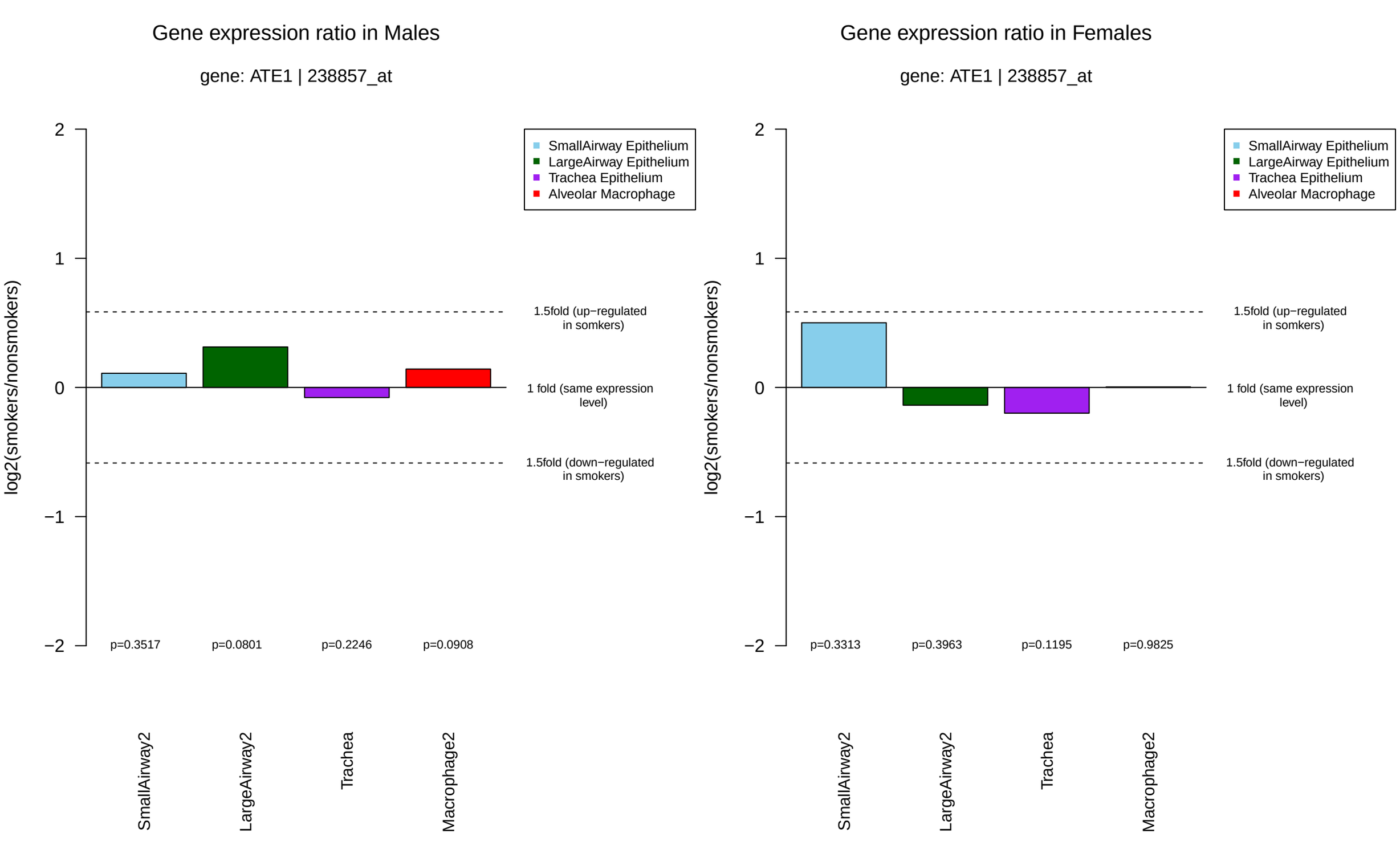
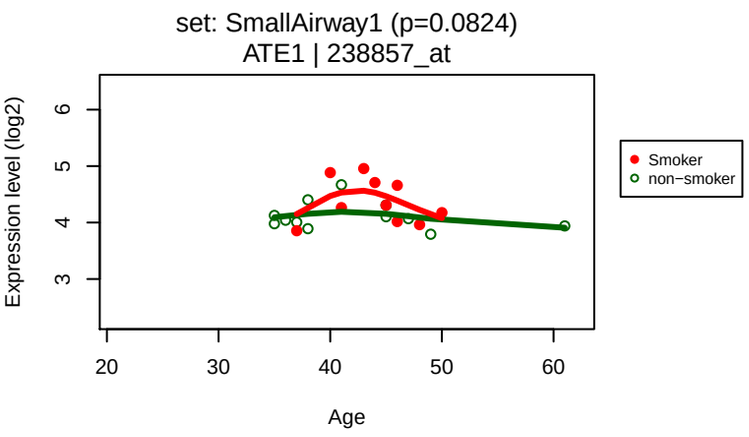
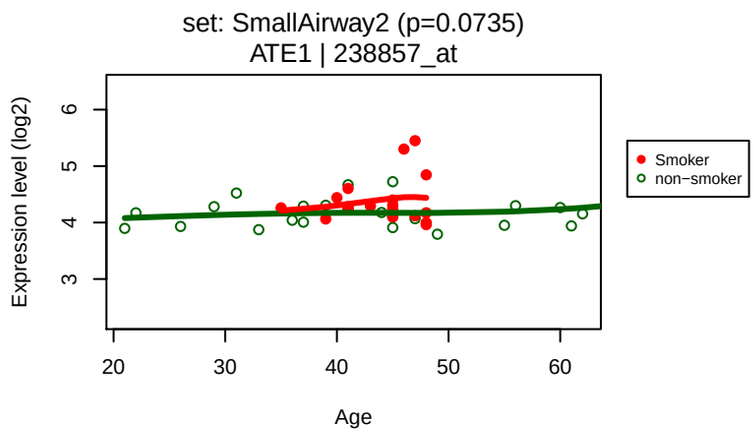
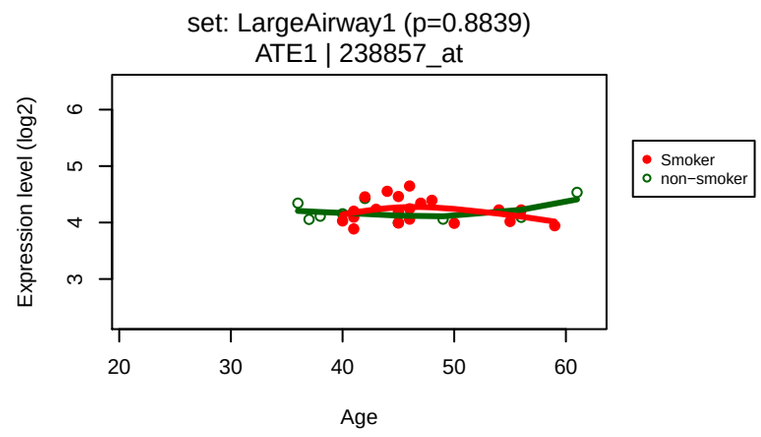

Gene expression of smokers and nonsmokers
gene: ATE1 | 238857_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0824
p=0.0735
p=0.0652
p=0.8839
p=0.1455
p=0.095
p=0.9795
p=0.1947
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

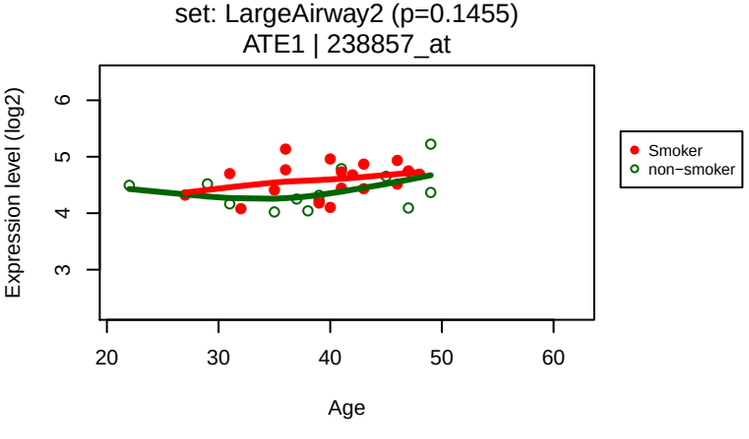

Gene expression ratio (smokers/nonsmokers)
gene: ATE1 | 238857_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0824
p=0.0735
p=0.0652
p=0.8839
p=0.1455
p=0.095
p=0.9795
p=0.1947
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

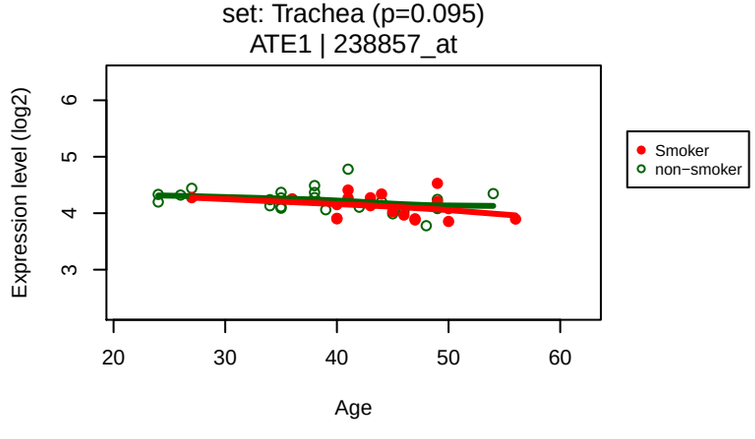

Gene expression ratio in Males
gene: ATE1 | 238857_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3517
p=0.0801
p=0.2246
p=0.0908
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATE1 | 238857_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3313
p=0.3963
p=0.1195
p=0.9825
SmallAirway2
LargeAirway2
Trachea
Macrophage2

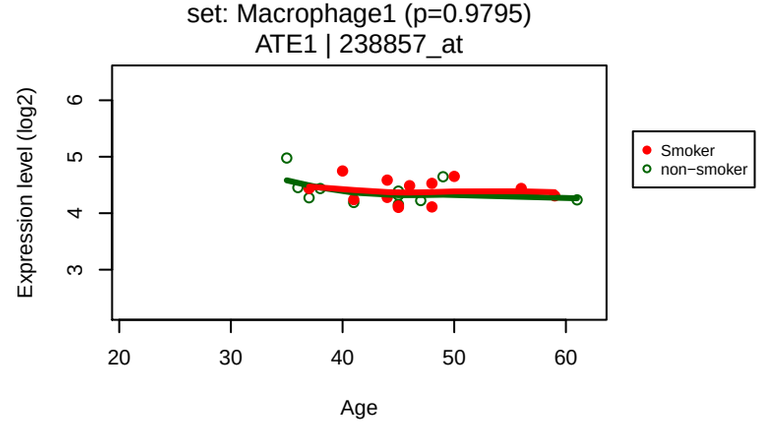

set: SmallAirway1 (p=0.0824)
ATE1 | 238857_at
Expression level (log2)
Age
Smoker
non−smoker

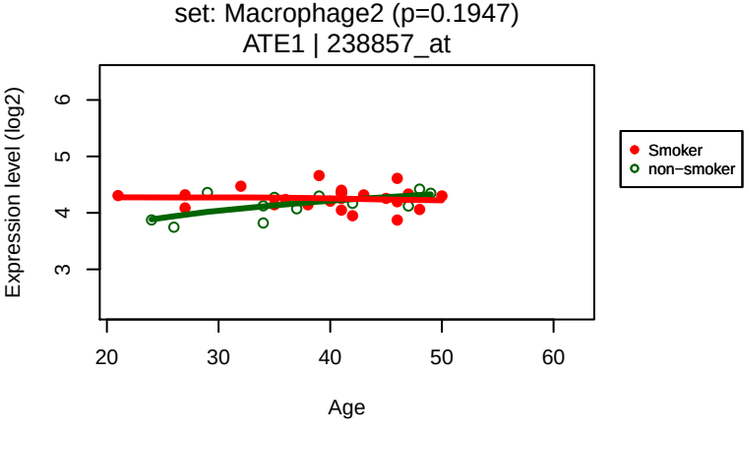

set: SmallAirway2 (p=0.0735)
ATE1 | 238857_at
Expression level (log2)
Age
Smoker
non−smoker

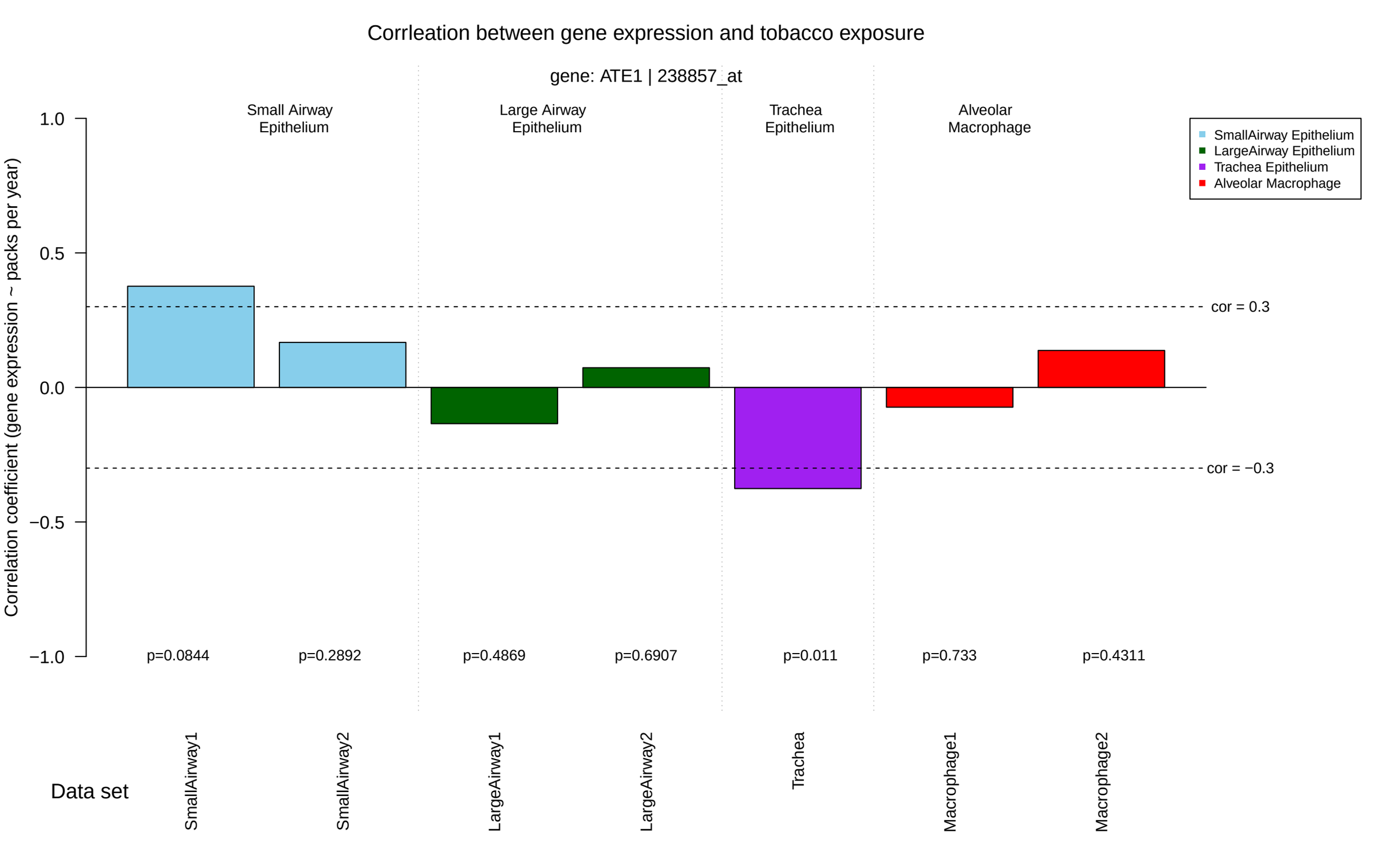

set: LargeAirway1 (p=0.8839)
ATE1 | 238857_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1455)
ATE1 | 238857_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.095)
ATE1 | 238857_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9795)
ATE1 | 238857_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1947)
ATE1 | 238857_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATE1 | 238857_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0844
p=0.2892
p=0.4869
p=0.6907
p=0.011
p=0.733
p=0.4311
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2